

Dr.-Ing. Christian Herff

Associate Professor

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🌐 neuralinterfacinglab.github.io

Education

- 2012–2016 **Dr.-Ing. Informatik**, *University of Bremen, Germany, Summa cum laude*
Speech Processes for Brain-Computer Interfaces
- 2006–2011 **Diploma Computer Science (MSc equivalent)**, *Karlsruhe Institute of Technology, Deutschland, 1.1*
Best 10% of Graduates
- 2011 **Master thesis**, *Institute for Infocomm Research, Singapore, 1.0*
Speech-related activations in the Brain, Funded by Scholarship of Foundation of German Business (SDW)
- 2009–2010 **Bachelor thesis**, *Indian Institute of Technology, Delhi, India*
Professor Naveen Garg, Funded by Scholarship of Foundation of German Business (SDW)

Work Experience

- Since 2025 **Associate Professor**, *Head of the Neural Interfacing Lab, Faculty for Health, Medicine and Life Science, Maastricht University, The Netherlands*
- 2022–2025 **Tenured Assistant Professor**, *Head of the Neural Interfacing Lab, Faculty for Health, Medicine and Life Science, Maastricht University, The Netherlands*
- 2019–2022 **Assistant Professor**, *Faculty for Health, Medicine and Life Science, Maastricht University, The Netherlands*
- 2018–2019 **Postdoctoral Researcher**, *Department of Neurosurgery, School for Mental Health & Neuroscience, Maastricht University, The Netherlands*
- Nov. 2018 **Visiting Scholar**, *University of Bergamo, Bergamo, Italy*
- Nov. 2017 **Visiting Scholar**, *ASPEN LAB, Old Dominion University, Norfolk, VA, USA*
- 2016–2018 **Postdoctoral Researcher**, *Cognitive Systems Lab, Universität Bremen, Germany*
- 2012–2016 **PhD student**, *Cognitive Systems Lab, University of Bremen, Germany, Started at Karlsruhe Institute of Technology, but lab moved to Bremen.*
- 2012 **Visiting Researcher**, *Schalk Lab, New York State Department of Health, Albany, NY, USA*
Funding by International Excellence Fund, Karlsruhe Institute of Technology

Funding and Awards

- 2024 Co-PI in NWO Take-Off: MOVE4Ward (Funding: 40,000€)
- 2024 KNAW: Early Career Award (Funding: 15,000€)
- 2023 IEEE SMC: Early Career Travel Award

- 2022 Kavli Foundation: 'NWB for Brain-Computer Interfaces' (Funding: 10,000€)
- 2020 Google Cloud Platform: 'End-to-End Speech Synthesis from Intracranial EEG' (Funding: 5,000€)
- 2020 Center for Integrative Neuroscience: 'Memory Bias In Decision-making (MEMOBID)' (Funding: 100,000€)
- 2019 NWO VENI Personal Grant: 'DEcoding Speech in SEEG (DEISIS)' (Funding: 249,600€)
- 2018 Postdoc-Initiative Plus: 'Feasibility Study for sEEG-based BCI' (Funding: 9,600€)
- 2018 3rd Price winner BCI award (1,000€)
- 2016 NSF Student travel award für das BCI Meeting 2016 (Funding: 1,100€)
- 2015 Price for best teaching in Computer Science at Karlsruhe Institute of Technology
- 2015 2nd price in Brain-Computer Interface competition at NER'2015 (300€)
- 2013 International Excellence Fund des Karlsruhe Institute of Technology (Funding: 10,000€)
- 2012 Feasibility Study of Young Scientists (Funding: 20,000€)
- 2006-2011 Scholarship of the Foundation of German Business

Selected Invited Talks

- 2025 Invited Speaker, 'Neurophysiology data for Brain-Computer Interface ', Medtech Day Kempenhaeghe, 20.11.2025
- 2025 Keynote Speaker, 'Speech BCI from brain-wide recordings ', NeuroDecoder, Paris, 9.9.2025
- 2025 Invited Speaker, 'Ideomotor principles to enhance interactive brain-environment interfacing', Universität Bielefeld, 28.4.2025
- 2025 Invited Speaker, 'Speech BCI', Basque Center for Language Technology, 6.3.2025
- 2024 Keynote Speaker, 'Speech BCIs from minimally invasive recordings', BCI Summer School, Utrecht, 18.9.2024
- 2024 Invited Speaker, 'Real-time Brain-Computer Interfacing from Intracranial EEG', KNF Dag, Lunteren, 23.5.2024
- 2023 Invited Speaker, 'Decoding Speech from Invasive Brain Signals', CuttingEEG, Frankfurt, 18.10.2023
- 2023 Invited Speaker, 'Decoding Higher-Order Cognition from sEEG', Donders Cognition, Brain, and Technology summer school, Nijmegen, 13.9.2023
- 2023 Invited Speaker, 'Speech BCI', Telluride Neuromorphic Cognition Engineering Workshop, Telluride, 3.7.2023
- 2023 Keynote Lecture, 'Decoding Higher-Order Cognition from Invasive Brain Signals', GTEC BCI Master Class, 19.6.2023
- 2022 Panelist: FutureBody Gathering, Berlin, 20.7.2022
- 2022 Panelist: Academic Careers at Neuromatch Academy, 20.7.2022

- 2022 Keynote Lecture 'Decoding Higher-Order Cognition from Invasive Brain Signals' at Annual Meeting of Institute of Neurological Sciences, Hacettepe University, Turkey, 7.6.2022
- 2022 Panelist: "CogHear Seminar on Imagined Speech", 7.4.2022
- 2022 Invited Lecture: "Decoding Speech from invasive EEG", Copenhagen University, Denmark 11.3.2022
- 2022 Invited Speaker, "Decoding Speech from invasive EEG", BrainCom Workshop 26.1.2022
- 2022 Invited Speaker, "Decoding Higher Order Cognition from invasive Brain Signals, University of Bremen, 24.1.2022
- 2021 Expert for Roundtable on Neurotechnology, United Nations, 23.7.2021
- 2021 Inaugural Speaker: "Machine Learning for Brain-Computer Interfaces' in BCI Society Next generation events, 21.1.2021
- 2020 Keynote Lecture 'Decoding Speech from Invasive EEG' at BCI Un-Conference, 23.7.2020
- 2018 Invited Speaker 'Brain-To-Speech' at 'Thought-to-Speech, Step 2'-Summit, Boston, 31.7.2018
- 2018 Invited Speaker 'Machine Learning on Brain Data' at Inaugural Meeting of Bremen City AI group, 31.5.2018
- 2018 Invited Speaker 'BCI – an overview' at 'International Cerebral Palsy Technology Summit', San Francisco, 3.-4.5.2018
- 2014–2017 5 more invited talks

Service to the Community

- Seit 2024 Section Head 'functional Neurosurgery'
- Seit 2023 Board Member 'BCI Society'

Conference Organization

- 2024 BCI Society — Chen Institute Joint BCI Meeting December 6-7, 2024, Shanghai, China

Session Organisation

- 2025 Session Chair 'Translating Speech Neuroprostheses', BCI Meeting, Banff, Canada
- 2024 Session Chair 'Speech Neuroprostheses', SfN, Chicago
- 2023 Session Organizer 'Speech Neuroprosthesis', BCI Meeting, Brussels
- 2023 Session Organizer 'Neural Representations for BCI', IEEE SMC, Oahu
- 2021 Session Chair 'Speech in Intracranial Recordings', SfN, Online
- 2021 Session Organizer 'Towards Speech Neuroprosthesis', BCI Meeting, Online
- 2020 Session Organizer 'Neural Signals for Spoken Communication', Interspeech, Online
- 2019 Tutorial Organizer: 'Biosignal-based speech processing: from silent speech to brain-computer interfaces', Interspeech, Graz

- 2019 Member of the programm committee: 'real-time functional imaging and neurofeed-back conference', Aachen-Maastricht
- 2018 Session Organizer 'BCI and Augmented/Virtual Reality', BCI Meeting, Asilomar
- 2018 Session Organizer 'Progress in Decoding Speech Processes using Intracranial Signals', BCI Meeting, Asilomar
- 2016 Session Organizer 'Decoding Speech Processes using Intracranial Signals', BCI Meeting, Asilomar
- 2014 Session Chair and Organizer 'Single-Trial Analysis of fNIRS data', IEEE EMBC, Chicago

Editor

- 2021 Guest Editor: PloS Digital Medicine
- Since 2020 Associate Editor: Frontiers in Neurotech
- 2018 Frontiers Research Topic Editor: Brain-Computer Interfaces and Augmented/Virtual Reality
- 2017 Frontiers Research Topic Editor: Detection and Estimation of Working Memory States and Cognitive Functions Based on Neurophysiological Measures

Reviewer (selected)

Nature, Neuron, Nature Communications, Nature Machine Intelligence, Nature Reviews Neuroscience, IEEE Transactions on Audio, Speech and Language Processing, IEEE Signal Processing Letters, IEEE Transactions on Biomedical Engineering, IEEE Transactions on Neural Systems and Rehabilitation Engineering, NeuroImage, Journal of Neural Engineering, Frontiers in Neuroscience, Interspeech, IEEE EMBC, IEEE SMC

Media Coverage

Television

- 2025 **TV Globo:** Fantastico: Paciente com paralisia volta a se comunicar só com a força do pensamento , 31.9.2025
- 2021 **RTV:** Een laptop die je gedachten leest, 5.10.2021
- 2017 **ARD Tagesschau:** Experte zu Facebook's BCI Plänen, 20.4.2017
- 2017 **Arte Xenius:** Hirnströme: Wie lässt sich die Kraft der Gedanken nutzen?, Report on 'Brain-to-text' study, 18.3.2017
- 2015 **3Sat Nano:** Befreiung aus dem eigenen Körper - Spracherkennung für Locked-In-Patienten, Report on 'Brain-to-text' study, 16.09.2015

Radio Interviews

- 2021 **BNR:** COMPUTER VERTAALT VRIJWEL DIRECT EEN WOORD DAT IEMAND DENKT NAAR SPRAAK, October 2021
- 2019 **Deutschlandfunk Kultur:** Interview, July 2019
- 2017 **1Live:** Interview, April 2017
- 2015 **NDR:** Interview, July 2015

- 2015 **RBB:** Die Profis, July 2015
- 2015 **ORF:** Wissen aktuell - Aus Hirnströmen lesen, was ein Mensch sagt, 18.06.2015
- 2015 **SWR2:** Sprechendes Gehirn: Das Brain-to-Text-System, 26.06.2015
- [Selected Print Media](#)
- 2025 **New York Times:** For Some Patients, the 'Inner Voice' May Soon Be Audible, 14.8.2025
- 2025 **Nature:** China pours money into brain chips that give paralysed people more control, 4.7.2025
- 2025 **Science:** This brain implant lets a man who lost his speech to ALS produce natural-sounding sentences instantaneously, 11.6.2025
- 2025 **Nature:** World first: brain implant lets man speak with expression — and sing, 11.6.2025
- 2025 **Nature:** Brain implant translates thoughts to speech in an instant, 31.3.2025
- 2024 **Scientific American:** Brain-to-Speech Tech Good Enough for Everyday Use Debuts in a Man with ALS, 14.8.2024
- 2024 **Nature:** Brain-reading device is best yet at decoding 'internal speech', 13.5.2024
- 2024 **Nature:** Mind-reading devices are revealing the brain's secrets, 20.2.2024
- 2023 **Nature:** Brain-reading devices allow paralysed people to talk using their thoughts, 23.08.2023
- 2023 **El País:** Un decodificador lee los pensamientos grabados por un escáner cerebral, 01.05.2023
- 2022 **NRC:** Computer leert gedachtenlezen: denk 'a' of 'u' en hij weet dat, 21.01.2022
- 2021 **Wired:** The Long Search for a Computer That Speaks Your Mind, 09.11.2021
- 2021 **New Scientist:** Gedachtelezende computer ontwikkeld voor mensen met spraakstoornissen, 18.10.2021
- 2021 **Washinton Post:** His voice silenced for years, a man can now communicate using only the electrical impulses from his brain, 16.7.2021
- 2021 **Science:** Brain signals 'speak' for person with paralysis, 16.7.2021
- 2020 **Kijk Magazine:** A.I. zet hersensignalen om in tekst, 02.04.2020
- 2020 **The Guardian:** Scientists develop AI that can turn brain activity into text, 30.03.2020
- 2019 **The Guardian:** Mind-reading tech? How private companies could gain access to our brains, 24.10.2019
- 2019 **de Volkskrant:** De computer die zegt wat je denkt – ook als je niet kunt bewegen, 28.6.2019
- 2019 **National Geographic:** New device translates brain activity into speech. Here's how. 24.4.2019
- 2019 **Wired:** The Machine That Reads Your Mind (Kinda) And Talks (Sorta), 24.4.2019
- 2019 **Science:** Artificial intelligence turns brain activity into speech, 2.1.2019
- 2018 **Plugged In: How Mind Machine Interfaces Will Transform the World** Scientific Advisor to Andrew Mangan, 13.03.2018

- 2017 **Quartz:** Facebook thinks the most useful digital assistant is the one that can read minds, 10.10.2017
- 2015 **Wired:** Brain-to-text tech lets computers read your thoughts, 19.06.2015
- 2015 **Zeit Online:** Forscher rekonstruieren Sprache aus Hirnströmen, 16.06.2015
- 2015 **SRF:** Sprache aus Hirnströmen erkennen, 16.06.2015

References

- **Prof. Dean Krusienski, PhD**
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Victoria Commonwealth University, VA, USA
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- **Prof. Cuntai Guan, PhD**
Full Professor
School of Computer Science and Engineering
College of Engineering
Nanyan Technological University, Singapore
Email: ctguan@ntu.edu.sg
Phone: (+65)6790 6205

Teaching

Coordination and Development, Maastricht University

- Since 2023 **Program Developer:** *New Bachelor in Brain Sciences*
- Since 2021 **Program Developer:** *New Bachelor in Regenerative Medicine and Technology*
- Since 2020 **Developer:** *New Master Health & Digital Transformation*
- Since 2019 **Course Planning Member:** *Master Neuromodulation*

Maastricht University

- SS **Mentor:** *Grant Writing Course*
- SS **Docent:** *Machine Learning for Health and Digital Transformation*
- SS **Docent:** *Department of Data Science: Research Project*
- WS **Docent:** *Closed-Loop Neural Interfaces*
- WS **Docent:** *Neurological Neuroscience*

University of Bremen

- SS2018 **Docent:** *Fundamentals of Machine Learning*
- SS2018 **Docent:** *Practical Course 'Brain Pattern Recognition'*
- SS2017 **Docent:** *Practical Course 'Brain Pattern Recognition'*
- SS2017 **Docent:** *Fundamentals of Machine Learning*
- WS2016 **Co-Docent:** *Biosignals and User Interfaces*
- WS2015 **Docent:** *Practical Course 'Brain Pattern Recognition'*

Karlsruhe Institute of Technology

- WS2014 **Docent:** *Practical Course 'Brain Pattern Recognition'*
- SS2014 **Docent:** *Brain-Computer Interfaces*
- SS2014 **Co-Docent:** *Biosignals and User Interfaces*
- WS2013 **Docent:** *Practical Course 'Brain Pattern Recognition'*
- SS2013 **Co-Docent:** *Biosignals and User Interfaces*
- WS2012 **Docent:** *Practical Course Biosignals*

Supervision

current PhD students

- Since 2024 Paul Weger, Maastricht University
- Since 2023 Sophia Gimple, Maastricht University
- Since 2023 Maria de Araújo Vitória, Maastricht University
- Since 2022 Joaquin Vega, Gran Sasso Science Institute

Former PhD students

- 2021-2025 Sophocles Goulis, On the role of the human hippocampus in economic decision making and framing: evidence from intracranial EEG and ultra-high field 7T fMRI during a monetary framing task, Maastricht University
- 2021-2025 Maxime Verwoert, On the Neural Symphony of Speech - Decoding Speech Production with Stereo-electroencephalography, Maastricht University
- 2019-2024 Maarten Ottenhof, ON THE OBIQUITY OF MOVEMENT - A decoding perspective on widespread motor-related neural activity, Maastricht University
- 2020-2023 Carlos Valle Araya, Improving Brain-to-Text Models Utilizing Recurrent Neural Networks and Transfer Learning for Neuroprosthetic Speech, Pontificia Universidad Católica de Chile, Chile

Co-Supervised PhD students

- 2018-2021 Margot Heijmans, Track and treat Parkinson's disease using wearable sensors and MRI, Maastricht University
- 2018-2021 Jeroen Habets, Prediction and real-life monitoring of DBS motor response in Parkinson's disease, Maastricht University

Member of the thesis committee

- 2026 Aurélie de Borman, Speech brain-computer interface based on human intracranial recordings , KU Leuven, Belgium
- 2025 Lynn Le, Reconstructing visual experiences from the neural code with deep learning, Radboud University, Netherlands
- 2025 Mohamed Baha Ben Ticha, Deep framework for speech decoding from ECoG, National Institute of Health and Medical Research, France
- 2024 Sofiane Jhilal, Speech BCIs for locked-in patients, Institute Pasteur, France
- 2023 Kinkini Bhadra, Non-invasive speech BCI, University of Geneva, Switzerland
- 2023 Hannah Bernhard, In The Event of Memory, Maastricht University, the Netherlands
- 2022 Shizhe Wu, Improving brain computer-interface for syllable imagery, University of Geneva, Switzerland
- 2021 Ciaran Cooney, Paradigm and AI Design Choices for Decoding Imagined Speech from Electroencephalography, University of Ulster, Northern Ireland
- 2021 Amaia Benitez-Andonegui, The Good Placement: Information-driven choice of fNIRS optode location and its impact on brain-computer interface performance, Maastricht University,the Netherlands

2020 Leon Skottnik, Understanding and improving neurofeedback-guided self-regulation: On the neuropsychological mechanisms of neurofeedback across mental tasks and time, Maastricht University, the Netherlands

2020 Niels Disbergen, Music to the Brain: investigating auditory scene analysis with polyphonic music, Maastricht University, the Netherlands

[Research Assistants](#)

Since 2019 2 Research Assistants im 'DESI' Projekt

2015-2018 3 Research Assistants im 'RESPONSE' Projekt

2012-2015 4 Research Assistants in verschiedenen fNIRS Projekten

[Master- und Bachelor theses](#)

Since 2012 > 70 supervised theses